

OSBORNE V. LOPES

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EDUCATION

Northeastern University <i>Master of Science, Information Systems</i> Coursework: Big Data and Intelligent Analytics, Data Architecture and Business Intelligence, Data Science Methods and Tools	Dec 2025 <i>Boston, MA</i>
University of Mumbai <i>Bachelor of Technology, Information Technology</i> Coursework: Data Mining and Business Intelligence, Database Management, Cloud Computing Services	Jun 2021 <i>Mumbai, India</i>

EXPERIENCE

Saayam For All <i>AI Engineer</i> - Architected volunteer-requestor semantic matching using Qdrant vector store with optimized chunking strategies, evaluating LLM cost-performance trade-offs across model sizes to select optimal embedding and generation models, integrating REST APIs and webhooks via n8n to reduce manual operational effort by 35% - Monitored end-to-end AI pipeline observability tracking output quality, retrieval accuracy and latency metrics across agentic workflows, surfacing donor and volunteer engagement trends via QuickSight dashboards that improved platform adoption by 25%	Feb 2026 – Present <i>Remote, US</i>
Atlas SP <i>AI Engineer Co-op</i> - Built and deployed a production RAG-based chatbot using Azure AI Search vector indexes, LangChain and FastAPI, enabling natural language querying of live financial data for internal teams and reducing repetitive analyst requests by 30% - Designed normalized and dimensional data models in Snowflake and optimized Delta Lake queries to retrieve data only on market days, improving query performance by 20% and enabling cleaner portfolio analytics for executive reporting - Automated and monitored data pipelines using ADF, Control-M and Alteryx, processing millions of financial records across multiple sources and reducing manual effort by 10 hrs/week	Jan 2025 – Aug 2025 <i>New York, NY</i>
Ingram Micro <i>Analytics Engineer</i> - Designed star-schema semantic models in Power BI with advanced DAX measures to track inventory turnover, order fulfillment and supplier performance, earning trust of cross-functional logistics leadership through executive-level KPI reporting across 5+ business teams - Analyzed customer order and delivery performance metrics across 10+ enterprise clients via incremental DBT models and Looker dashboards, conducting A/B tests on reporting formats to identify SLA breach trends and reducing repeat complaints by 15% - Reduced inventory shortfalls by 20% through demand forecasting pipelines and delivered healthcare device logistics dashboards using QuickSight and Tableau, automating ingestion via Airflow within CI/CD pipelines to ensure SLA adherence across operations	Oct 2022 – Aug 2023 <i>Mumbai, India</i>
ICICI Securities <i>Data Analyst</i> - Partnered with trading and operations stakeholders to gather reporting requirements, delivering structured Tableau dashboards that saved 8 hours weekly and improved visibility into trade and portfolio data - Led 5 Disaster Recovery drills, resolved 20+ live issues during market hours and documented operational SOPs across NSE and BSE trading platforms, ensuring uninterrupted operations for thousands of active traders - Dived deep into trading datasets using Python, R and Teradata to build reusable SQL models using window functions and partition pruning, reducing processing time by 40% and owning end-to-end dataset reliability for reporting across 5+ business teams	Jun 2021 – Sep 2022 <i>Mumbai, India</i>

PROJECTS

RAG Job Assistant - Built a production-grade RAG pipeline indexing 5,000+ job postings into ChromaDB using HuggingFace sentence-transformer embeddings (all-MiniLM-L6-v2), enabling semantic search over job descriptions - Architected a multi-agent retrieval system using LangGraph with ReAct-pattern agents for query decomposition, semantic search and answer synthesis, integrating LangChain and Groq API (Llama 3.3 70B) with prompt engineering to manage context windows, multi-turn interactions and output quality evaluation	May 2025
NewsScope - Built an industry-specific news aggregator that scrapes, processes and surfaces real-time business news using Python (BeautifulSoup, Scrapy) and DBT - Designed an end-to-end ETL pipeline on GCP using Dataflow for processing, PineCone for vector embeddings and BigQuery as the data warehouse, enforcing data quality via Pydantic 2 and Pytest validations - Containerized and deployed the platform with Docker, exposing a Streamlit interface for browsing curated news by industry	Apr 2024

SKILLS

Languages: Python, Java, R, SQL, Pro*C Databases & Warehousing: Snowflake, MongoDB, DynamoDB, Redshift Cloud Platforms: Azure (ADF, Blob Storage, Synapse), AWS (S3, Glue, EMR, Lambda, Athena, Kinesis, IAM), GCP (BigQuery, DataFlow, GCS), Docker Tools: Airflow, DBT, Apache Spark, Kafka, Databricks, Alteryx, Control-M, Git Analytical Toolkit: Tableau, PowerBI, Looker, Google Data Studio, QuickSight, Excel AI & Automation: RAG, Agentic AI, LangChain, LangGraph, HuggingFace, FastAPI, ChromaDB, Qdrant, PineCone, LLM, n8n, Prompt Engineering, Claude API, Model Context Protocol (MCP)	
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